Lctr 14: Magnitude, Phase, Group Delay, and Ideal vs. Real Filters

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Focus: Sections 6.1–6.4 (Pages 423–456)

14.1 The Big Picture: What Magnitude and Phase Tell Us

Why This Matters

In Lectures 12–13 we developed the Fourier transform and its properties. We know that an LTI system's frequency response $H(j\omega)$ tells us how the system modifies each frequency component. But $H(j\omega)$ is a *complex-valued* function, carrying both a **magnitude** $|H(j\omega)|$ and a **phase** $\angle H(j\omega)$. So far we have focused mostly on magnitude ("how much of each frequency gets through?"). This lecture examines **both** magnitude and phase, introduces the concept of **group delay**, and then uses these tools to understand the fundamental properties and limitations of frequency-selective filters—both ideal and real.

Roadmap for this lecture:

1. Magnitude–phase representation and the energy spectral density.
2. Why phase matters: the striking effect of scrambling phase.
3. Linear phase equals pure time delay; nonlinear phase causes distortion.
4. Group delay: the effective delay at each frequency.
5. Ideal frequency-selective filters: impulse response, causality, and realizability.
6. Nonideal filters: transition bands, ripple, and the bandwidth–rise-time trade-off.

14.2 Magnitude–Phase Representation

14.2.1 Polar Form of the Fourier Transform

Any Fourier transform $X(j\omega)$ can be written in polar form:

$$X(j\omega) = |X(j\omega)|\, e^{j\angle X(j\omega)}$$

where:

- $|X(j\omega)|$ is the **magnitude spectrum** (always ≥ 0).
- $\angle X(j\omega)$ is the **phase spectrum** (in radians).

For real signals, conjugate symmetry (Lecture 13) tells us:

- $|X(j\omega)|$ is an **even** function of ω.
- $\angle X(j\omega)$ is an **odd** function of ω.

14.2.2 Energy Spectral Density

From Parseval's relation (Lecture 13):

$$\underbrace{\int_{-\infty}^{+\infty} |x(t)|^2 \, dt}_{\text{total energy}} = \frac{1}{2\pi} \int_{-\infty}^{+\infty} |X(j\omega)|^2 \, d\omega$$

The quantity $|X(j\omega)|^2$ is the **energy spectral density**: it tells us how the signal's total energy is distributed across frequencies. The energy in a frequency band $[\omega_1, \omega_2]$ is:

$$E_{[\omega_1,\omega_2]} = \frac{1}{2\pi} \int_{\omega_1}^{\omega_2} |X(j\omega)|^2 \, d\omega + \frac{1}{2\pi} \int_{-\omega_2}^{-\omega_1} |X(j\omega)|^2 \, d\omega$$

14.2.3 Log-Magnitude and the Decibel Scale

In practice, magnitudes span many orders of magnitude, so we often use a **logarithmic** scale. The standard unit is the **decibel** (dB):

$$\boxed{|H(j\omega)|_{\text{dB}} = 20 \log_{10} |H(j\omega)|}$$

$\lvert H \rvert$	dB value	Interpretation
1	0 dB	No change
$1/\sqrt{2} \approx 0.707$	-3 dB	Half-power point (cutoff)
0.1	-20 dB	Attenuated to 10%
0.01	-40 dB	Attenuated to 1%
10	$+20$ dB	Amplified $10\times$

Key Insight

The -3 dB point is the standard definition of **cutoff frequency**: it is where $|H|^2 = 1/2$, meaning the signal power is halved. Every additional -20 dB corresponds to another factor-of-10 attenuation. You will use the dB scale constantly in filter design and Bode plots (Lecture 15).

14.3 Why Phase Matters

Why This Matters

Students often focus exclusively on magnitude and neglect phase. This is a serious mistake. The phase spectrum carries critical information about the **structure** and **timing** of a signal. Two signals with the same magnitude spectrum but different phase spectra can look completely different in the time domain.

14.3.1 The Phase-Scrambling Thought Experiment

Imagine taking a signal $x(t)$, computing its Fourier transform $X(j\omega) = |X|\, e^{j\angle X}$, then replacing the phase with random values while keeping the magnitude intact. The inverse transform gives a signal with the same energy at every frequency, but a completely different waveform.

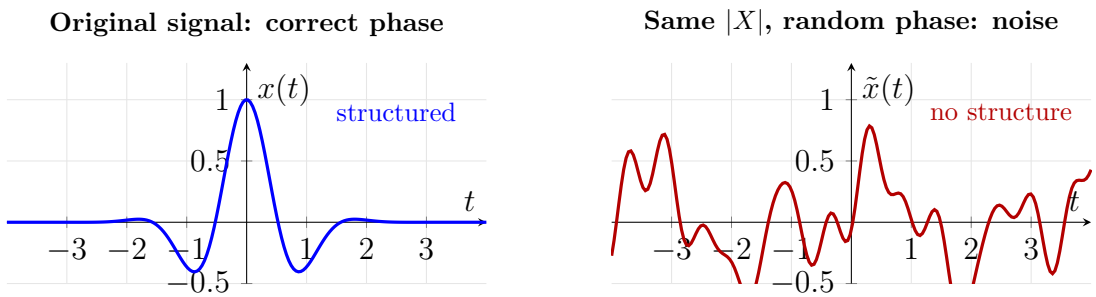

Figure 1: Both signals have exactly the same magnitude spectrum $|X(j\omega)|$ and therefore the same total energy. But the original (left) is a recognizable pulse, while the phase-scrambled version (right) looks like noise. Phase carries the structural information.

Key Insight

Phase carries structure; magnitude carries energy. If you had to throw away one, throwing away the magnitude and keeping the phase would preserve more of the signal's recognizable features (edges, transitions, timing). This is well known in image processing: the phase of an image's 2D Fourier transform encodes edges and shapes, while the magnitude encodes contrast and brightness.

14.4 Linear Phase and Time Delay

14.4.1 CT Linear Phase

Consider a system with frequency response:

$$H(j\omega) = |H(j\omega)|\, e^{-j\omega t_d}$$

where t_d is a constant. The phase is $\angle H(j\omega) = -\omega t_d$—a **linear function** of ω.

If the input is $x(t)$ with FT $X(j\omega)$, then the output is:

$$Y(j\omega) = |H(j\omega)|\, e^{-j\omega t_d}\, X(j\omega)$$

Case 1: Unity magnitude ($|H| = 1$ for all ω). Then $Y(j\omega) = e^{-j\omega t_d} X(j\omega)$, and by the time-shift property:

$$y(t) = x(t - t_d)$$

The output is a **perfect, undistorted copy** of the input, delayed by t_d. No frequency is treated differently—they all experience the same delay.

Case 2: Non-unity magnitude (e.g., a lowpass filter with linear phase). The output is a *filtered and delayed* version of the input. The filtering changes the amplitude of different frequencies, but because the phase is linear, all frequency components experience the same delay t_d. The waveform shape is modified (by the magnitude response) but not *distorted by phase misalignment*.

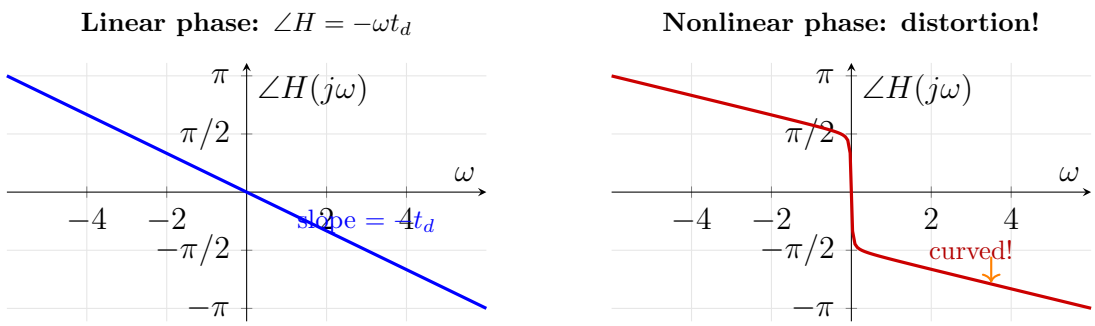

Figure 2: Left: Linear phase is a straight line through the origin. All frequencies experience the same delay. Right: Nonlinear phase means different frequencies are delayed by different amounts, causing waveform distortion even if the magnitude response is flat.

14.4.2 DT Linear Phase

In discrete time, the same principle holds. If $H(e^{j\omega})$ has phase $\angle H(e^{j\omega}) = -\omega n_d$ for integer n_d, then:

$$y[n] = x[n - n_d] \quad \text{(pure sample delay of } n_d \text{ samples)}$$

If n_d is not an integer, the system still represents a "fractional delay"—a more subtle concept that requires interpolation between samples.

14.4.3 Why Nonlinear Phase Causes Distortion

Visual Intuition

Think of a signal as a collection of sinusoids at different frequencies. If all sinusoids are delayed by the same amount (linear phase), they arrive "in step" and the waveform is preserved. If different frequencies experience different delays (nonlinear phase), the sinusoids arrive out of step and the waveform is distorted—even though the magnitude of each component is unchanged.

Nonlinear Phase Distortion of a Two-Component Signal

Consider $x(t) = \cos(\omega_1 t) + \cos(\omega_2 t)$ with $\omega_1 = 1$ and $\omega_2 = 3$.
System A (linear phase): $\angle H = -2\omega$. Both components are delayed by $t_d = 2$:

$$y_A(t) = \cos(\omega_1(t-2)) + \cos(\omega_2(t-2)) = \cos(t-2) + \cos(3(t-2))$$

The output is an exact delayed copy of the input.
System B (nonlinear phase): $\angle H(\omega_1) = -2$, $\angle H(\omega_2) = -9$. The component at ω_1 is delayed by $2/1 = 2$ s, but the component at ω_2 is delayed by $9/3 = 3$ s:

$$y_B(t) = \cos(t-2) + \cos(3t-9) = \cos(t-2) + \cos(3(t-3))$$

The two components arrive at *different* times. The waveform shape changes—this is phase distortion.

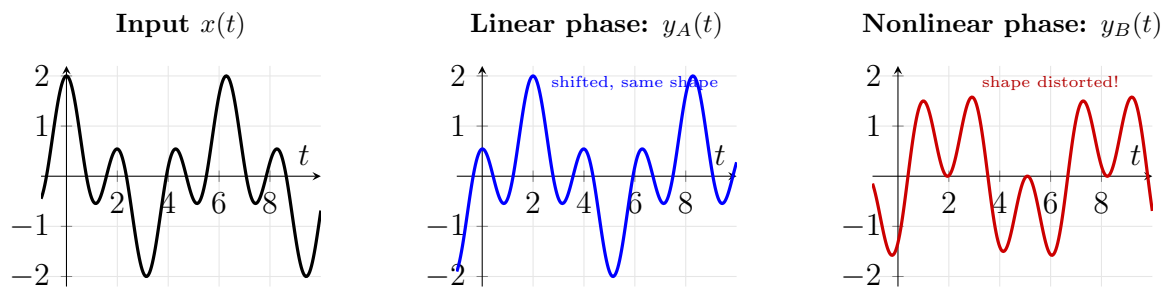

Figure 3: The input (left) passes through two systems with the same magnitude response but different phase. System A (linear phase, center) produces a perfect delayed replica. System B (nonlinear phase, right) distorts the waveform because the two frequency components are delayed by different amounts.

14.5 Group Delay

14.5.1 Definition

The **group delay** of a system is defined as:

$$\boxed{\tau(\omega) = -\frac{d}{d\omega}\big[\angle H(j\omega)\big]}$$

It measures the **effective delay experienced by a narrow band of frequencies centered at** ω. For a system with perfectly linear phase, $\angle H = -\omega t_d$, so $\tau(\omega) = t_d$ (constant for all ω). For a system with nonlinear phase, $\tau(\omega)$ varies with frequency—different frequency bands experience different delays.

14.5.2 Narrowband Interpretation

Consider a **narrowband** signal: a sinusoidal carrier modulated by a slowly varying envelope. The *carrier* propagates at the rate determined by the phase at the carrier frequency. The *envelope* propagates at the rate determined by the group delay at the carrier frequency. When the group delay is not constant, the envelope of a narrowband signal is distorted—it spreads out or compresses differently at different carrier frequencies.

Computing Group Delay for a First-Order System

Consider the first-order lowpass system $H(j\omega) = \frac{1}{1+j\omega/\omega_c}$ (from Lecture 11, with $\omega_c = 1/RC$).
Step 1: The phase is $\angle H(j\omega) = -\arctan(\omega/\omega_c)$.
Step 2: Differentiate:

$$\tau(\omega) = -\frac{d}{d\omega}\left[-\arctan(\omega/\omega_c)\right] = \frac{1/\omega_c}{1+(\omega/\omega_c)^2} = \frac{\omega_c}{\omega_c^2+\omega^2}$$

Step 3: Evaluate numerically for $\omega_c = 3$ rad/s:

- At $\omega = 0$: $\tau = 1/3 \approx 0.333$ s (maximum delay)
- At $\omega = 3$ $(= \omega_c)$: $\tau = 3/18 = 1/6 \approx 0.167$ s
- At $\omega = 10$: $\tau = 3/109 \approx 0.028$ s (nearly zero delay)

The group delay is *not* constant—this system has nonlinear phase and will cause some distortion of wideband signals.

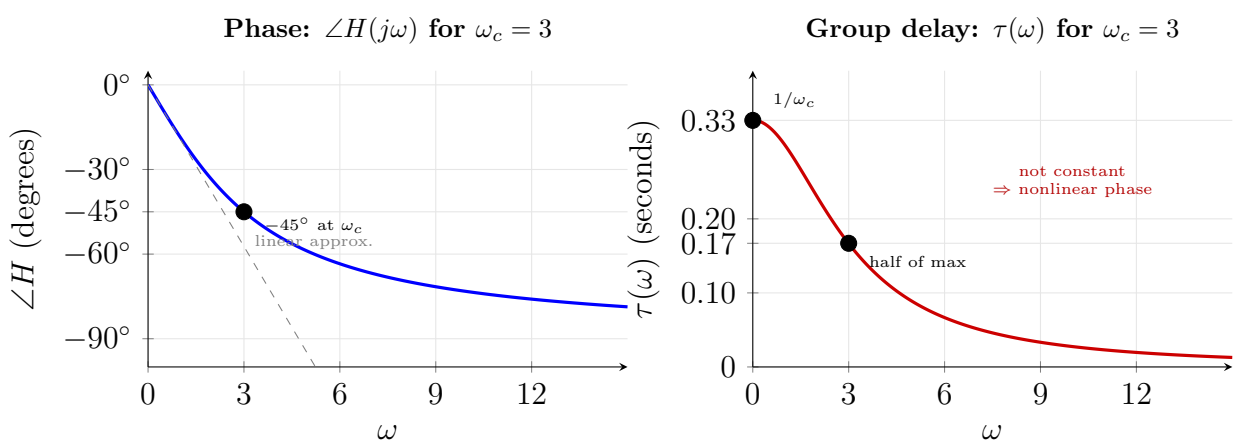

Figure 4: Left: The phase of a first-order system is $-\arctan(\omega/\omega_c)$—clearly nonlinear (not a straight line). Right: The corresponding group delay $\tau(\omega)$ decreases from $1/\omega_c$ at DC to nearly zero at high frequencies. Low-frequency components are delayed more than high-frequency ones.

Key Insight

Summary of the delay story:

- **Linear phase** ($\angle H = -\omega t_d$) $\Rightarrow \tau(\omega) = t_d$ (constant) $\Rightarrow$ no phase distortion.
- **Nonlinear phase** $\Rightarrow \tau(\omega)$ varies with $\omega \Rightarrow$ phase distortion.
- For many applications (audio, communications), having approximately constant group delay over the passband is as important as having a flat magnitude response.

14.6 Ideal Frequency-Selective Filters

Why This Matters

In Lecture 11 we introduced filter vocabulary (lowpass, highpass, bandpass). In Lecture 13, duality showed us that a rectangular spectrum corresponds to a sinc impulse response. Now we put these pieces together: the **ideal lowpass filter** has a perfectly rectangular magnitude response and linear phase—and we can compute its impulse response, step response, and discover a fundamental limitation: it is **not physically realizable**.

14.6.1 The Ideal CT Lowpass Filter

The ideal lowpass filter passes all frequencies below ω_c and blocks everything above:

$$H_{\text{LP}}(j\omega) = \begin{cases} e^{-j\omega t_d}, & |\omega| \le \omega_c \\ 0, & |\omega| > \omega_c \end{cases}$$

The $e^{-j\omega t_d}$ factor gives the filter linear phase (constant delay t_d). The magnitude is:

$$|H_{\text{LP}}(j\omega)| = \begin{cases} 1, & |\omega| \le \omega_c \\ 0, & |\omega| > \omega_c \end{cases}$$

14.6.2 Impulse Response of the Ideal Lowpass Filter

Taking the inverse Fourier transform:

$$h(t) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{-j\omega t_d}\, e^{j\omega t}\, d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega(t-t_d)}\, d\omega$$

Evaluating:

$$\boxed{h(t) = \frac{\sin\big(\omega_c(t-t_d)\big)}{\pi(t-t_d)} = \frac{\omega_c}{\pi}\,\text{sinc}\left(\frac{\omega_c(t-t_d)}{\pi}\right)}$$

This is a **sinc function** centered at $t = t_d$.

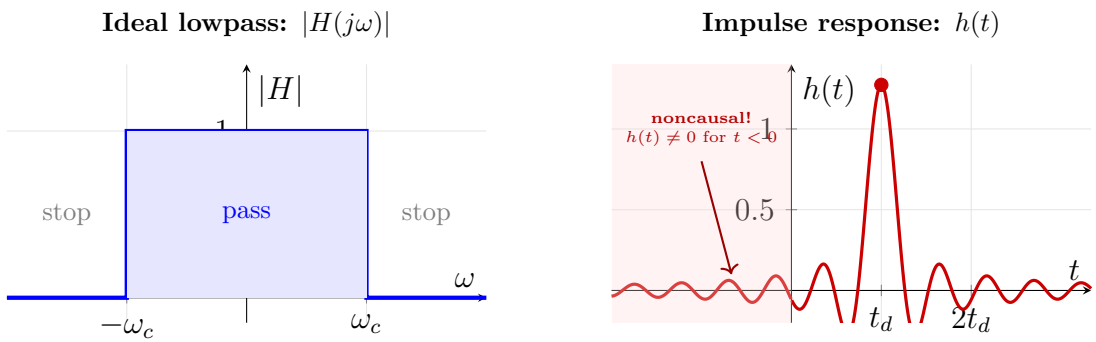

Figure 5: The ideal lowpass filter (left) has a sinc impulse response (right). The sinc extends to $t = -\infty$, meaning the system requires knowledge of *future* input values. This makes it **noncausal** and **physically unrealizable**.

14.6.3 Why Ideal Filters Are Unrealizable

The sinc impulse response is **noncausal**: $h(t) \neq 0$ for $t < 0$. This means the system would need to know the input *before it arrives*—which is physically impossible for a real-time system.

Additionally, the sinc function extends infinitely in both directions and decays only as $1/t$, meaning it is barely absolutely integrable. These properties create fundamental limitations.

Key Insight

No physically realizable filter can have a perfectly sharp frequency cutoff. This is a direct consequence of the time–frequency trade-off: a rectangular shape in one domain requires an infinitely extended sinc in the other. All real filters must have a *gradual* transition between passband and stopband.

14.6.4 Step Response of the Ideal Lowpass Filter

The step response $s(t) = \int_{-\infty}^{t} h(\tau)\, d\tau$ of the ideal lowpass filter is the integral of the sinc function. This results in oscillations around the step edge—a manifestation of the **Gibbs phenomenon** we first saw in Lecture 10.

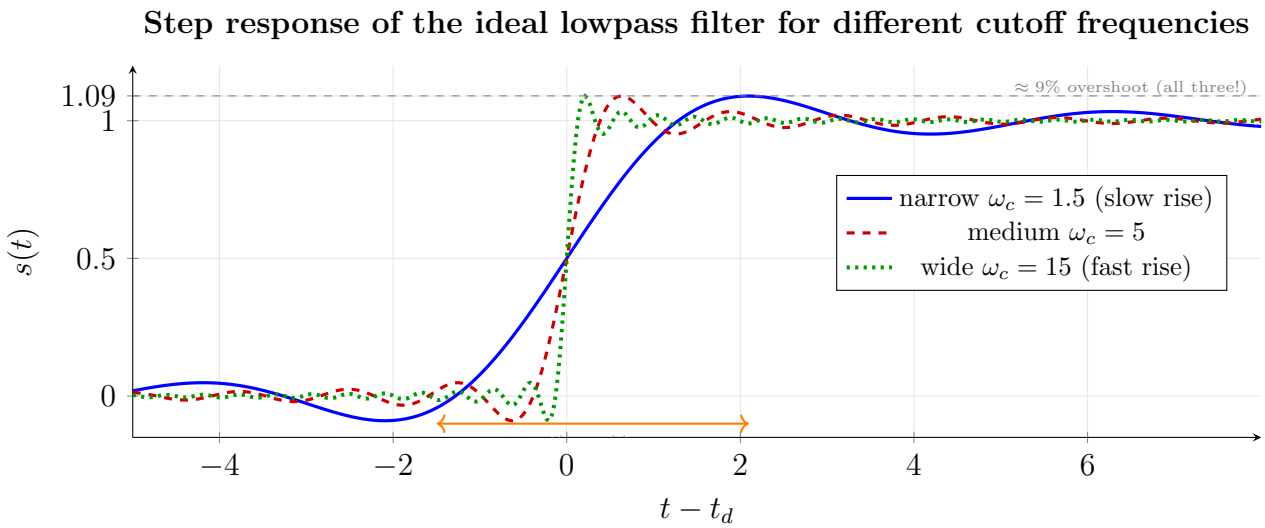

Figure 6: Step response of ideal lowpass filters with different bandwidths. Wider $\omega_c \Rightarrow$ faster rise but the $\approx 9\%$ overshoot (Gibbs phenomenon) persists. The rise time t_r is inversely proportional to ω_c.

14.7 Nonideal Filters: Real-World Specifications

Since ideal filters are unrealizable, we work with **nonideal** approximations. A real filter is characterized by several practical specifications. The key idea is that a real filter does *not* have a single "cutoff frequency" with a perfectly sharp transition. Instead, there is a **range of frequencies** over which the filter transitions from passing to blocking—and we need multiple parameters to describe this transition.

14.7.1 From Ideal Cutoff to Practical Specifications

For an ideal filter, there is one number: the cutoff frequency ω_c. For a real filter, this single boundary is replaced by **two** frequencies:

- ω_p (**passband edge**): the highest frequency at which the filter still meets its passband specification (magnitude close to 1, within the allowed ripple).
- ω_s (**stopband edge**): the lowest frequency at which the filter meets its stopband specification (magnitude below the required attenuation level).

Between ω_p and ω_s lies the **transition band**—the region where the filter is "rolling off" but has not yet reached full attenuation. The ideal filter has $\omega_p = \omega_s = \omega_c$ (zero-width transition), which is unrealizable.

The traditional -3 dB cutoff frequency ω_c (where $|H| = 1/\sqrt{2}$) typically falls somewhere inside the transition band, often close to ω_p. In Butterworth filter design, ω_c is defined as the -3 dB point and coincides with ω_p.

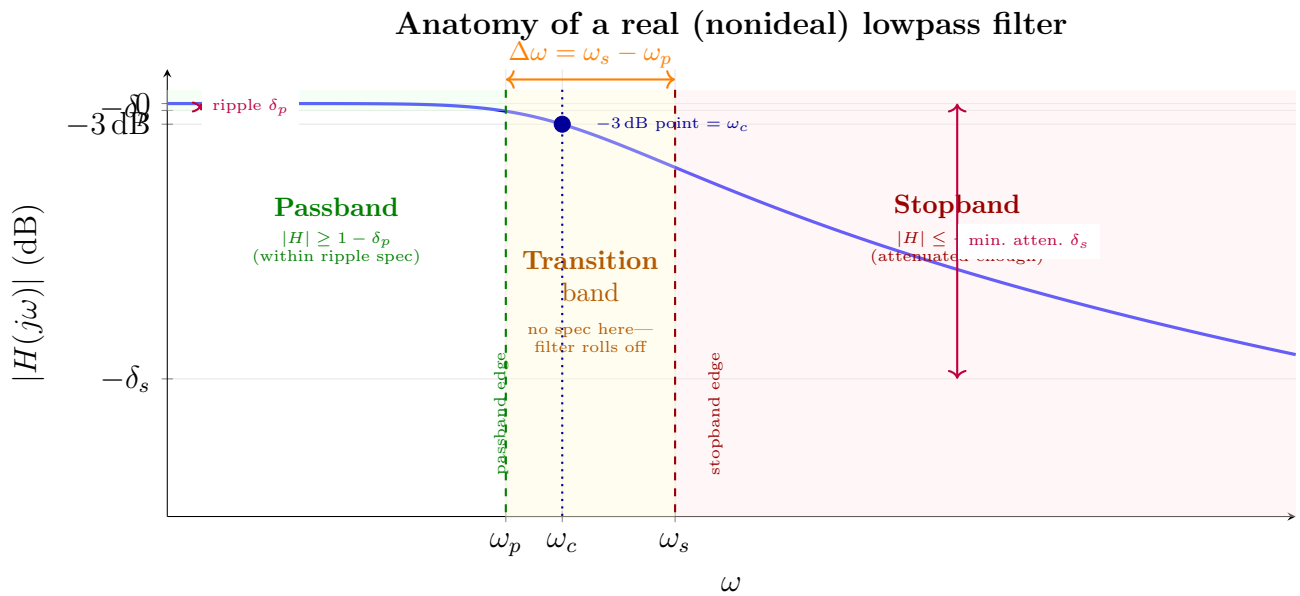

Figure 7: A real lowpass filter has a passband (magnitude within δ_p of unity), a stopband (attenuation $\geq \delta_s$ dB), and a **transition band** between them. The classical -3 dB cutoff frequency ω_c falls inside the transition band. The ideal filter has $\omega_p = \omega_s$ (zero-width transition), which is unrealizable.

14.7.2 Summary of Specification Parameters

Key specification parameters:

- **Passband edge frequency** ω_p: the highest frequency guaranteed to meet the passband requirement.
- **Stopband edge frequency** ω_s: the lowest frequency guaranteed to meet the stopband requirement.
- **Cutoff frequency** ω_c: traditionally defined as the -3 dB point (where $|H|^2 = 1/2$). For Butterworth filters, $\omega_c \approx \omega_p$. For Chebyshev or elliptic filters, ω_c may differ from both ω_p and ω_s.
- **Transition bandwidth** $\Delta\omega = \omega_s - \omega_p$: the width of the roll-off region. Smaller $\Delta\omega$ means a sharper filter but requires higher filter order (more complexity).
- **Passband ripple** δ_p: maximum deviation from unity gain in the passband (in dB). Typical values: 0.5–3 dB.
- **Stopband attenuation** δ_s: minimum attenuation in the stopband (in dB). Typical values: 20–60 dB.

14.7.2 The Fundamental Trade-off: Bandwidth vs. Rise Time

There is an approximate relationship between the bandwidth ω_c (or $\Delta\omega$) and the rise time t_r of a filter's step response:

$$\boxed{\omega_c \cdot t_r \approx \text{constant} \quad (\text{typically } \approx \pi \text{ to } 2\pi)}$$

What this means:

- A **wider** passband (larger ω_c) gives a **faster** step response (smaller t_r).
- A **narrower** passband gives a **slower** response.
- A **sharper** transition ($\omega_s \approx \omega_p$, small $\Delta\omega$) requires a **longer** impulse response $\Rightarrow$ more ringing in the time domain.

This is the same time–frequency trade-off we have seen throughout the course, now stated as a precise engineering design constraint.

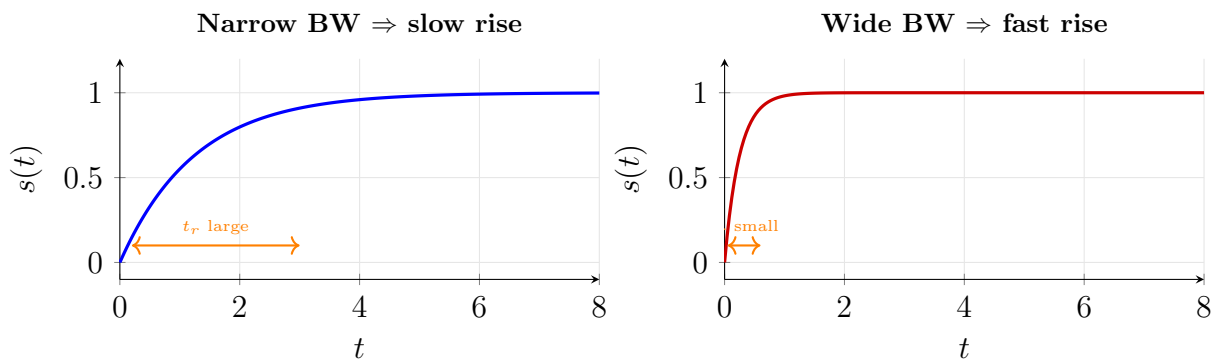

Figure 8: The bandwidth–rise-time trade-off. A narrow-bandwidth filter (left) responds slowly. A wide-bandwidth filter (right) responds quickly. You cannot have both sharp frequency selectivity and fast time-domain response.

14.7.3 Sharper Transition vs. More Ringing

Besides the rise-time trade-off, attempting to make the transition band narrower (closer to the ideal sharp cutoff) introduces more **oscillatory ringing** in the time-domain response. This is directly related to the Gibbs phenomenon from Lecture 10: the sharper the frequency-domain cutoff, the more pronounced the sidelobes of the sinc-like impulse response.

Key Insight

The three-way design trade-off for filters:

1. **Transition width** $\Delta\omega$: how sharp is the cutoff?
2. **Time-domain ringing**: how much overshoot/oscillation is acceptable?
3. **Filter order/complexity**: how many components or coefficients are needed?

Improving any one of these metrics typically worsens at least one of the others. Filter design is the art of finding the best compromise for a given application.

14.8 DT Ideal and Nonideal Filters

Everything we said about CT filters has a DT parallel.

14.8.1 The DT Ideal Lowpass Filter

$$H_{\mathrm{LP}}(e^{j\omega}) = \begin{cases} e^{-j\omega n_d}, & |\omega| \le \omega_c \\ 0, & \omega_c < |\omega| \le \pi \end{cases} \qquad \text{(periodic with period } 2\pi\text{)}$$

Its impulse response is:

$$h[n] = \frac{\sin\big(\omega_c(n - n_d)\big)}{\pi(n - n_d)}$$

This is a **discrete-time sinc**, and it shares the same problem: $h[n] \neq 0$ for $n < 0$ (noncausal), so the ideal DT lowpass is also unrealizable. The same trade-offs between transition bandwidth, ringing, and filter length apply.

14.8.2 Practical DT Filters

In discrete time, we approximate the ideal filter with a **finite-length** impulse response (FIR filter) by truncating and windowing the sinc, or with an **infinite** impulse response (IIR filter) using recursive structures. These were introduced conceptually in Lecture 11 and will be analyzed quantitatively in Lecture 15.

14.9 Summary: The Complete Frequency-Domain Picture

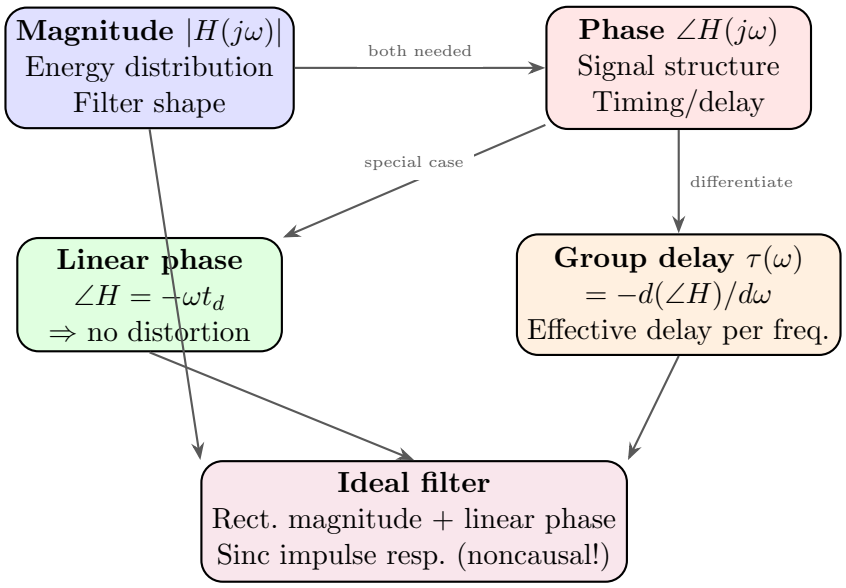

Figure 9: Concept map for Lecture 14. Magnitude controls energy and filter shape; phase controls structure and delay. Group delay quantifies the frequency-dependent delay. The ideal filter combines rectangular magnitude with linear phase, producing a noncausal sinc impulse response.

	Ideal Filter	**Real (Nonideal) Filter**
Magnitude	Perfectly rectangular	Gradual roll-off
Phase	Perfectly linear	Approximately linear in passband
Transition band	Zero width	Nonzero width $\Delta\omega$
Impulse response	Sinc (infinite, noncausal)	Finite or decaying (causal)
Realizable?	**No**	**Yes**

14.10 Common Mistakes to Avoid

1. **Ignoring phase entirely:** Many students focus only on the magnitude response. Phase is equally important for signal fidelity. A system with flat magnitude but highly nonlinear phase will still distort the signal.

2. **Confusing group delay with phase delay:** The **phase delay** is $-\angle H(j\omega)/\omega$ (the delay of the carrier); the **group delay** is $-d(\angle H)/d\omega$ (the delay of the envelope). For linear phase they are the same, but in general they differ.

3. **Thinking ideal filters exist in practice:** No physical system can produce a perfectly sharp cutoff. Every statement about "ideal lowpass" is a mathematical idealization. Real design always involves trade-offs.

4. **Forgetting the sign in the group delay formula:** $\tau(\omega) = -d\angle H/d\omega$ (note the minus sign). A phase that becomes more negative with increasing ω (like $-\arctan(\omega)$) gives a *positive* group delay. Getting the sign wrong inverts the physical meaning.

5. **Using the -3 dB point for all filter types:** The -3 dB cutoff is standard for Butterworth-type filters, but other filter families (Chebyshev, elliptic) may define the passband edge differently (e.g., at the maximum ripple point). Always check the specification convention.

6. **Confusing the Gibbs phenomenon with filter imperfection:** The $\approx 9\%$ overshoot in the step response of an ideal lowpass filter is *not* a defect of a specific filter design—it is a fundamental consequence of the sharp frequency cutoff. Even the "perfect" ideal filter exhibits Gibbs oscillations.

Key Insight

The central message of this lecture: Magnitude and phase are equally important for characterizing signals and systems. Linear phase preserves waveform shape; group delay quantifies frequency-dependent delay. Ideal filters are mathematically clean but physically unrealizable—real filter design is about managing the trade-offs between transition sharpness, time-domain ringing, and complexity.

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